

# Assessing Harmful Algal Blooms in Mississinewa Lake: a Current Progress Report with 2025 Bloom Conditions

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## OVERVIEW

- **Monitoring water quality is crucial for understanding and addressing environmental and human health.**
- **Harmful Algal Blooms (HABs) are becoming more common and more intense.**
- **HABs can negatively impact ecosystems by contributing to eutrophication, and releasing harmful toxins.**
- **Goal is to generate a model using remote sensing to predict the occurrence of HABs in inland waters of the North-Central U.S.**



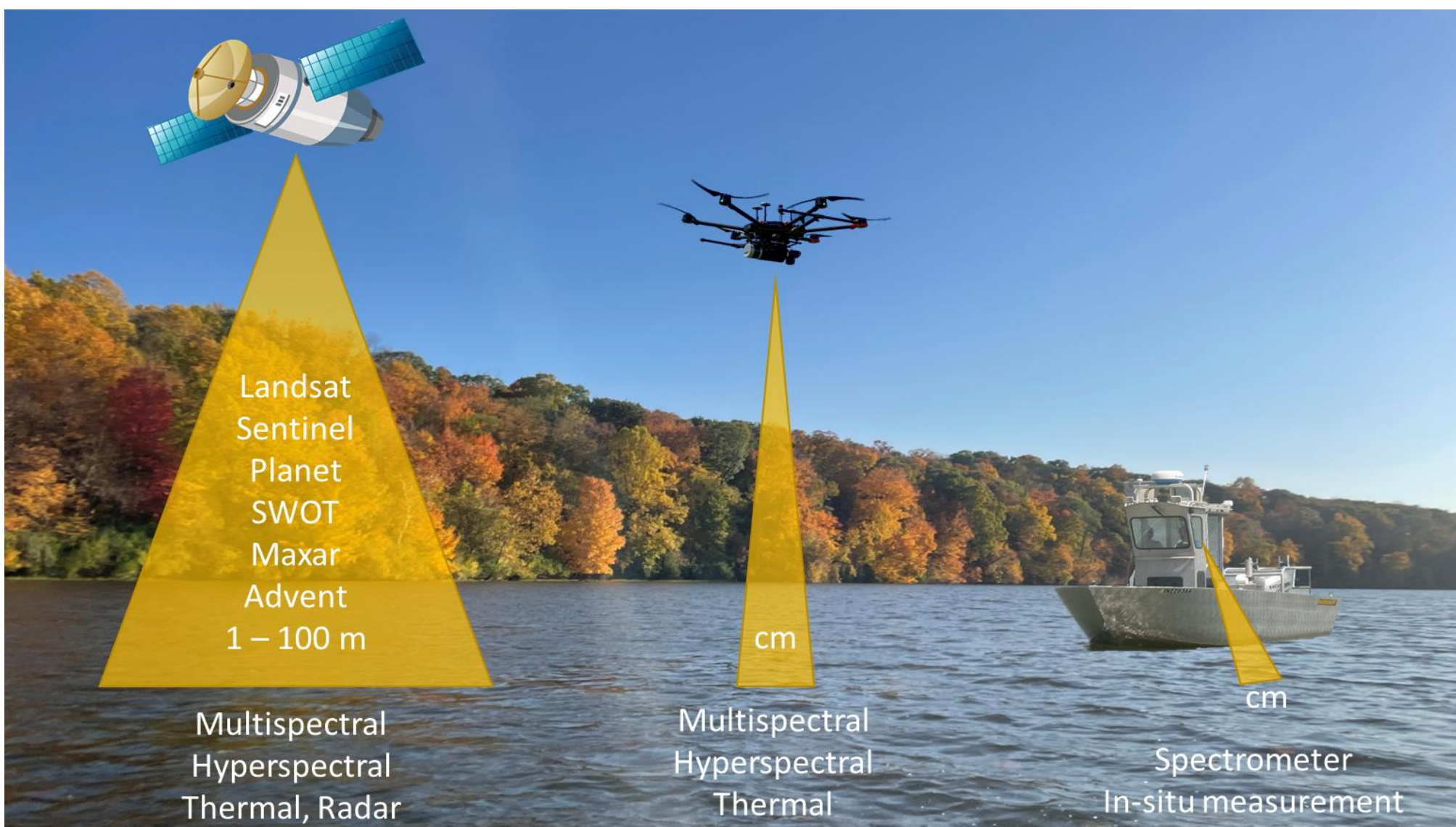
Left: Indiana with four sampled Lakes marked from left to right: Shafer, Freeman, Mississinewa, Salamonie)

## OBJECTIVES

1. Quantify the biogeochemical state of inland waters by analyzing their optical and thermal properties via remote sensing
2. Identify preconditions that contribute to HAB formation and predict the potential occurrence.

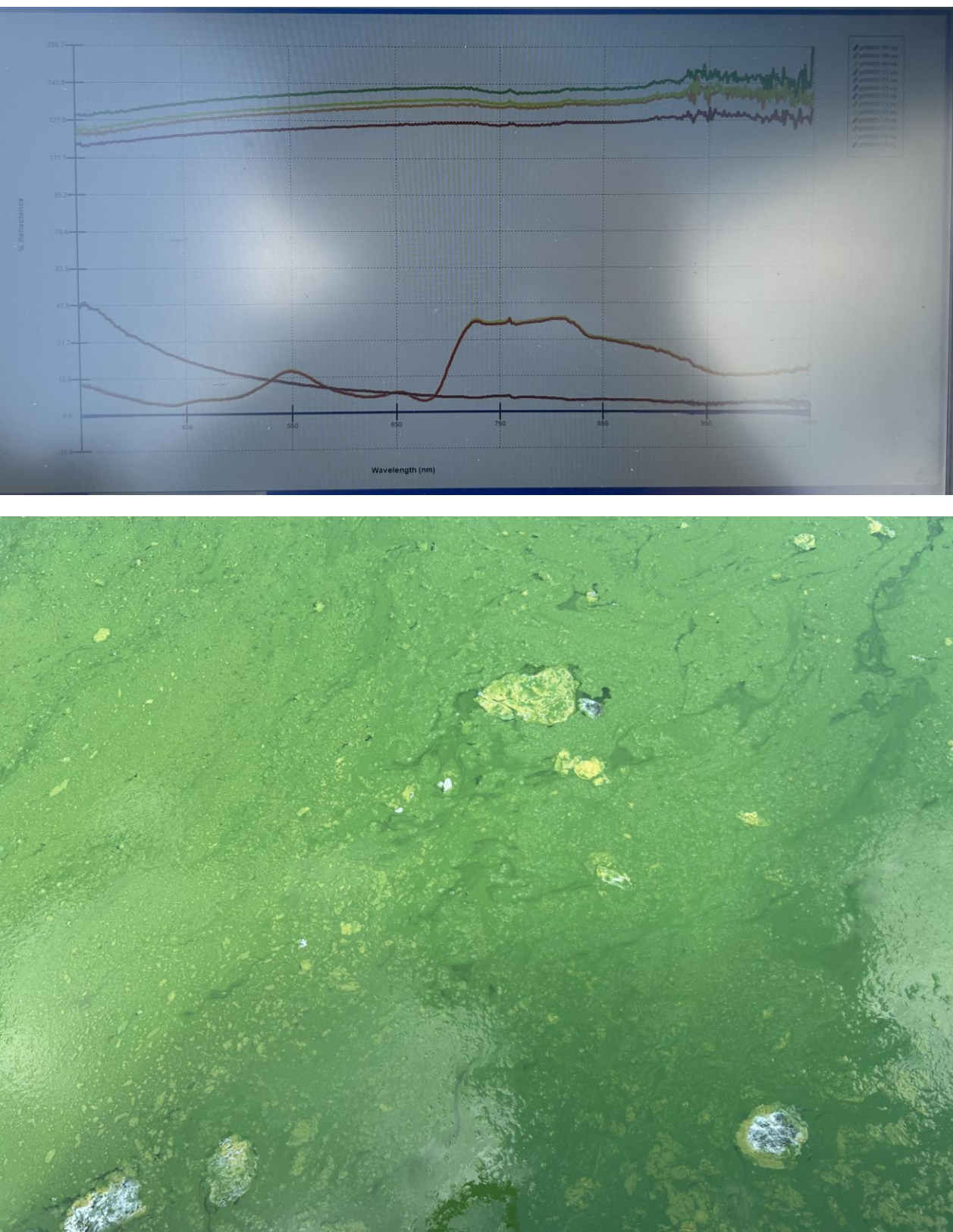
## SAMPLING METHODOLOGY

- In-situ **spectral reflectance** from water's surface recorded with a GER1500
- **Water temperature, ODO, conductivity or specific conductance, oxidation reduction potential, TDS or turbidity and pH** vertical profiles with a YSI multiparameter sonde
- Water samples collected undergo lab analysis
  - **Microcystin** with U.S. EPA Method 546
  - **Chlorophyll-a** with U.S. EPA Method 445.0
  - **Soluble nutrients** ammonia, nitrate-nitrite, and orthophosphate, and total phosphorus analyzed with U.S. EPA Methods 103, 114, 118365.1
- Imagery from remote sensing platforms collected during and evaluated post in-situ sampling.



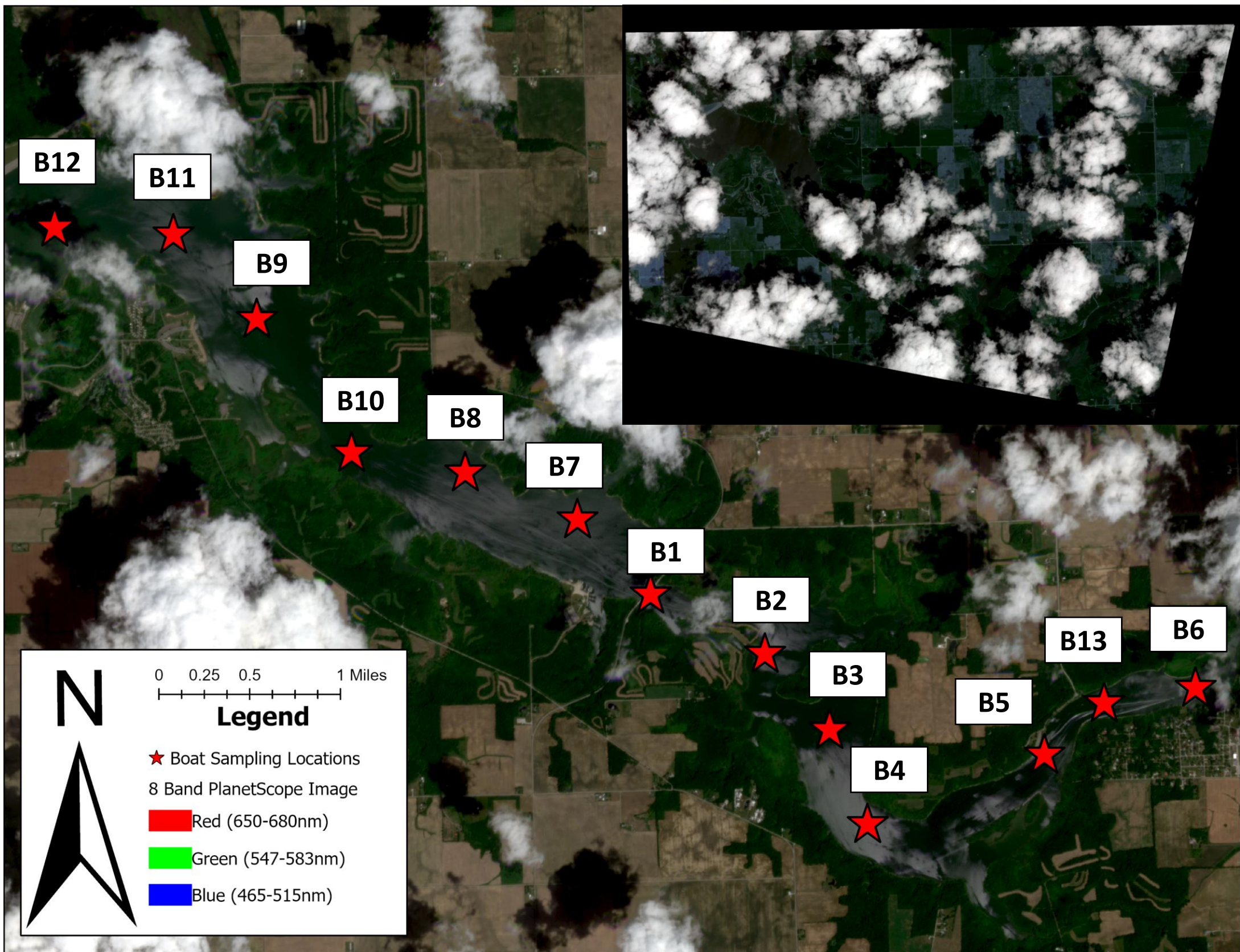
Data collection platforms and scale of sampling

## FIELD COLLECTION



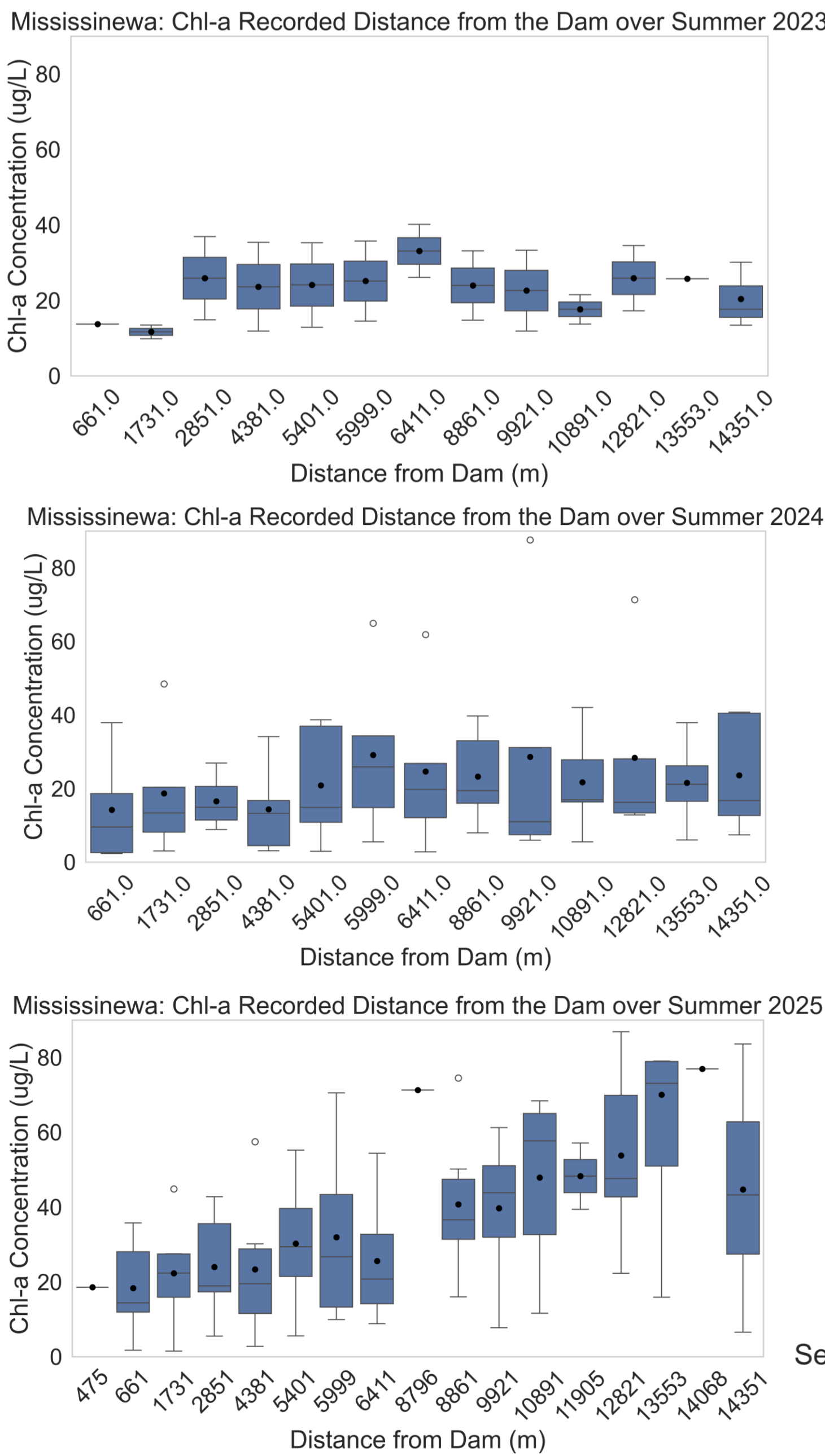
Top: Spectral radiance from the GER1500 handheld spectrometer over algal bloom, which is used for remote sensing calibration.

Bottom: 'Worst' algae bloom seen on Mississinewa this year, August 6, 2025.



Eight-band SuperDove PlanetScope Image for Lake Mississinewa 6/24/2024. Inset shows cloud cover from 7/18/24 for comparison.

## RESULTS

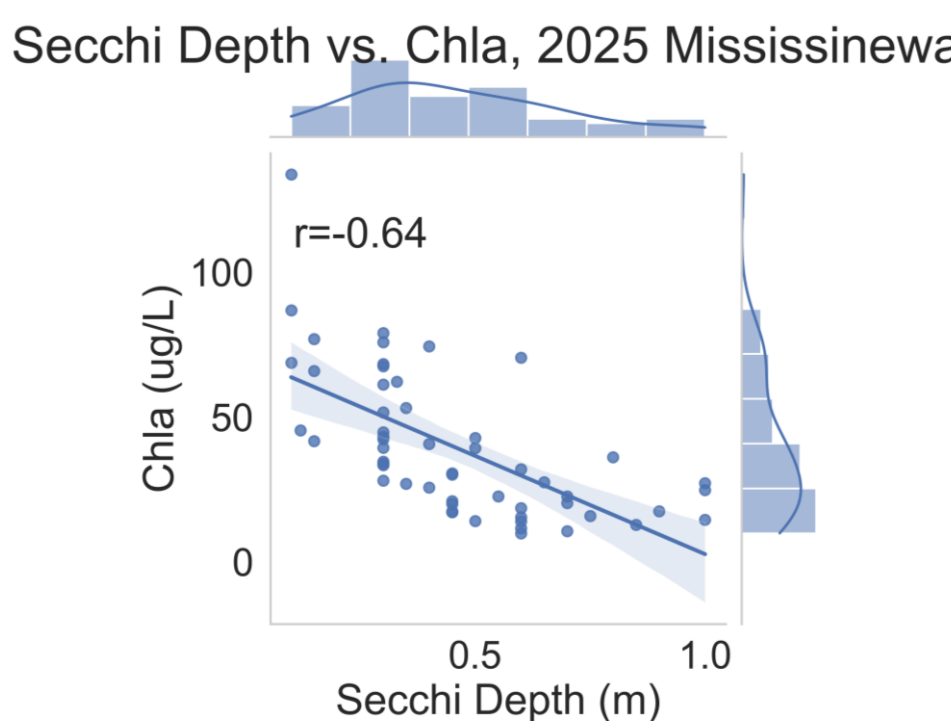


Right: 2025 Secchi Disk Depth vs, Chl-a Concentrations, illustrates a negative relationship between turbidity and Chl-a.

Boxplots of Chlorophyll-a concentrations by location for each year, 2023 (top) to 2025 (bottom), over the sampling season (May-Sep). This season had both the largest range of data and the highest concentrations.

This was due to the bloom conditions (see bottom left photo) not seen since 2019 based on IDEM records.

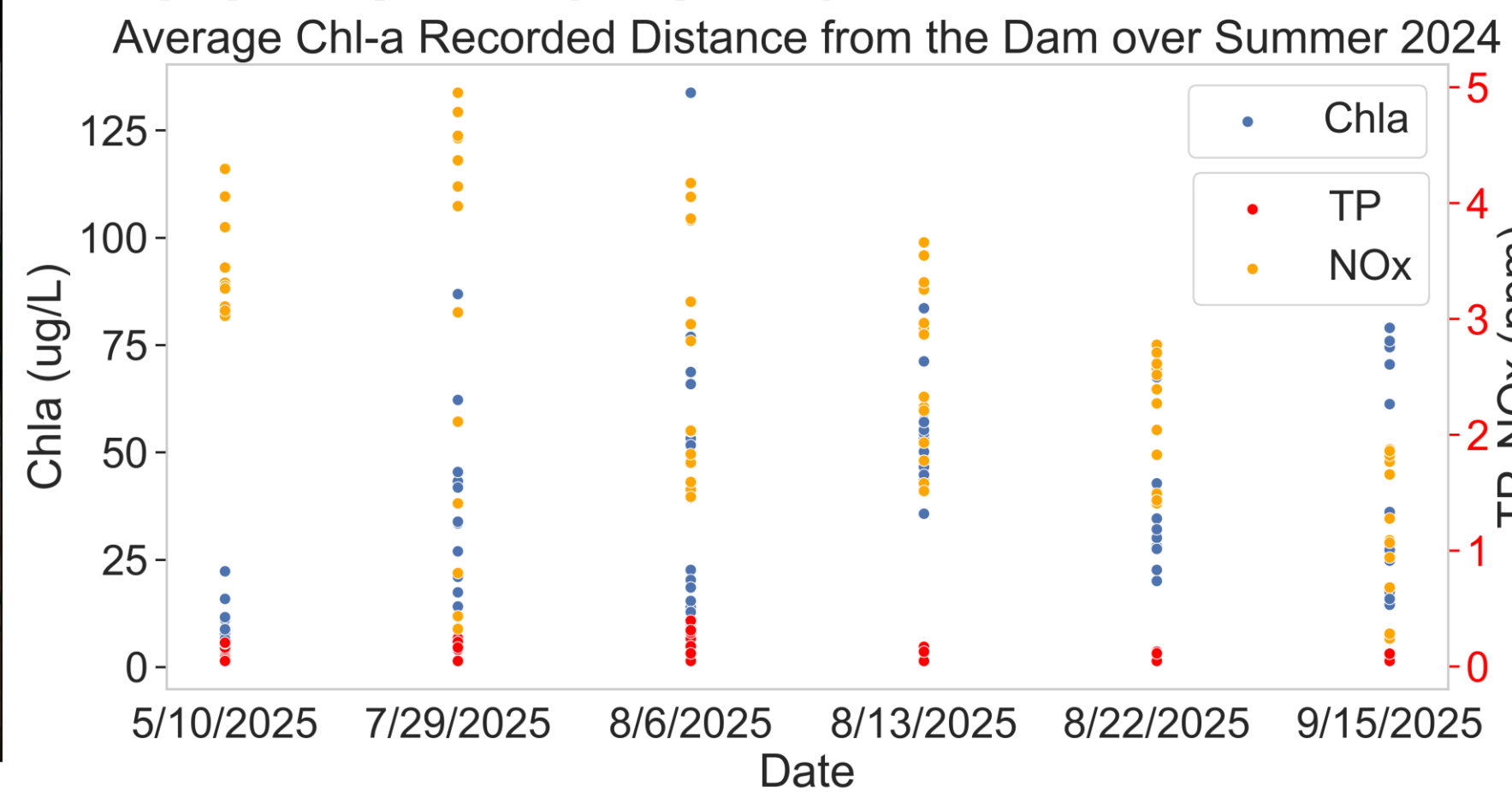
Differences in conditions that led to the blooms occurring 2025 but not the other years are being analyzed.



Thank you to everyone who has helped on this project and assisted with getting our data and processing samples!



## CONCLUSIONS



Above: 2025 Chl-a, Nitrate-Nitrite, and Total Phosphorous (Chl-a, NOx, and TP) by Date (points from all locations included)

- There appears to be an increase in TP and NOx corresponding with higher readings of Chl-a
- Lab analysis of water samples is nearly complete, after which analysis can focus on the role of temperature and pre-bloom environmental conditions that led to such high nutrient levels and intense bloom events in 2025.
- The Chl-a recorded this year was the **highest of all 3 years** and Mississinewa had the highest Microcystin in **all of Indiana in 2025**

## FUTURE WORK

- Utilize arcpy within ArcGIS to automate the extraction of remotely sensed imagery from sample sites.
- Analyze biogeochemical data paired with remotely sensed parameters to determine connections between biogeochemical state and HABs.
- Develop a predictive machine learning model for HAB risk using these data with supplementary information on land use and weather.

Special thanks to Mandy Limiac, Areesha Saleem and Max Ahrbecker.

Look at Figure 5 (for Chl-a peaks in spectra): Burket, M., Olmanson, L., & Brezonik, P. (2023). Comparison of Two Water Color Algorithms: Implications for the Remote Sensing of Water Bodies with Moderate to High CDOM or Chlorophyll Levels. *Sensors*, 23, 1071.

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